

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
2nd Semester of B. Pharm. Examination
University Theory Examination October/November 2017
PH120 & PH 128 Pharmaceutical Engineering-II

Date: 05/10/2017, Thursday Time: 10:00a.m. to 01:00 p.m. Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
 2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
 3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
 4. Figures to right indicate marks.
 5. Draw neat labeled sketches wherever necessary.
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SECTION - I

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|------------|--|-----------|
| Q 1 | Attempt all questions. Each question is of one mark. | 20 |
| 1. | Evaporation occurs only
A) after boiling
B) after extreme cooling
C) at surface of a liquid
D) if boiling occurs at atmospheric pressure | |
| 2. | Crystallization, evaporation and distillation is a mean of
A) separating soluble substances in solution
B) separating insoluble substances in solutions
C) separating filtrate from solution
D) all of these | |
| 3. | Distillate is a liquid that
A) mixes together
B) condenses after distillation
C) does not sink due to very fine particles
D) a substance that will not dissolve | |
| 4. | In fractional distillation, a large surface area for condensation is provided through
A) a thermometer
B) water bath
C) a fractionating column
D) reflux condenser | |
| 5. | Vacuum distillation mainly used in
A) Synthetic Industry
B) Chemical Industry | |

- C) Oil Industry
 - D) Food Industry
6. The moisture in the solid after attaining equilibrium is known as
 - A) Bound moisture
 - B) Unbound moisture
 - C) Equilibrium moisture
 - D) Free moisture
 7. The moisture inside the substance is known as _____
 - A) Bound moisture
 - B) Unbound moisture
 - C) Equilibrium moisture
 - D) Free moisture
 8. _____ is an instrument used to measure relative humidity
 - A) Barometer
 - B) Manometer
 - C) Pressure gauge
 - D) Psychrometer
 9. _____ temperature is the actual temperature of the air measured by an ordinary thermometer
 - A) Both Dry bulb and Wet bulb
 - B) Wet bulb
 - C) Dry bulb
 - D) None
 10. _____ is the actual mass of water vapor in one cubic foot of air
 - A) Absolute Humidity
 - B) Relative Humidity
 - C) Specific Humidity
 - D) Total Humidity
 11. Which of the following statements about the basic principle of sedimentation is False?
 - A) The denser a biological structure is, the faster it sediments in a centrifugal field.
 - B) The more massive a biological particle is, the slower it moves in a centrifugal field.
 - C) The denser the buffer system is, the slower the particle will move in a centrifugal field.
 - D) The greater the centrifugal force is, the faster the particle sediments.
 12. How solid sodium carbonate be obtained from sodium carbonate solution?
 - A) Centrifugation
 - B) Filtration
 - C) Evaporation
 - D) All of the above
 13. Process used to separate insoluble particles from liquids is
 - A) Drying
 - B) Filtration
 - C) Sieving
 - D) Extraction

14. Addition of filter aid in filtration resulting _____ rate of filtration.
 - A) Increase
 - B) Decrease
 - C) No change
 - D) No useful
15. Following is an example of industrial filters
 - A) Leaf filter,
 - B) Filter press,
 - C) Rotary filter
 - D) All of the above
16. Following is an example of industrial centrifuges
 - A) Perforated basket centrifuge
 - B) Sedimentation type centrifuge
 - C) Cartridge type centrifuge
 - D) (a) & (b) both
17. Duhring's Rule and Raoult's law principles applicable for
 - A) Drying
 - B) Evaporation
 - C) Distillation
 - D) All of the above
18. Absolute alcohol is prepared by one of the following method.
 - A) Azeotropic distillation
 - B) Simple distillation
 - C) Vacuum distillation
 - D) Steam distillation
19. _____ type of distillation technique is used for extracting most of the volatile oils such as clove, eucalyptus
 - A) Azeotropic distillation
 - B) Simple distillation
 - C) Flash distillation
 - D) Steam distillation
20. Increase in surface area in filtration resulting _____ rate of filtration.
 - A) Increase
 - B) Decrease
 - C) No change
 - D) No useful

SECTION – II

Q 2 Attempt any TWO of the following.

Differentiate the following.

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|---|--|-----------|
| A | (1) Ideal solutions with Real Solutions | 05 |
| | (2) Single effect with multiple effect evaporators | |
| B | Classify the evaporator. Briefly describe the factors affecting evaporation. | 05 |
| C | Give the Principle of Duhring's Rule and Raoult's law. List the factors affecting evaporation. | 05 |

- Q 3 Attempt any TWO of the following.**
- | | | |
|----------|--|-----------|
| A | Describe the various mechanisms of filtration process. | 05 |
| B | Briefly describe the factors affecting the process of filtration. | 05 |
| C | Give the construction, working and application with labeled diagram of filter press. | 05 |

SECTION – III

- Q 4 Attempt any FOUR of the following.**
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|----------|--|-----------|
| A | Define centrifugation. Write applications of centrifugation process in field of Pharmacy. | 05 |
| B | Describe principle, working, advantages, disadvantages and applications of Perforated basket centrifuge. | 05 |
| C | Write a detailed note on distillation column. | 05 |
| D | Classify the distillation methods. Discuss in detail about steam distillation. | 05 |
| E | Discuss McCabe Thiele method for calculation of number of theoretical plates. | 05 |
| F | Give construction and working of Fractional distillation with neat labeled diagram. | 05 |
- Q 5 Attempt any FOUR of the following.**
- | | | |
|----------|--|------------------------|
| A | Discuss in detail about Fluidized bed dryer. | 05 |
| B | Classify the dryer as per its mechanism. Briefly describe the terms Equilibrium Moisture content (EMC) and Free Moisture Content (FMC). | 05 |
| C | Describe the factors affecting selection of dryers. Classify the dryers. | 05 |
| D | If a mixture of air and water vapour at 80° F and 760 mm Hg pressure has humidity of 0.010 lb water vap/lb dry air, and if it saturated with water vapour at the same temperature and pressure the humidity is 0.0233 lb water vap/lb dry air. What will be the % humidity (H_p).
Give the salient features of Psychrometric chart. | 03
02 |
| E | Enlist the methods for measurement of humidity. Briefly describe wet bulb and dry bulb method | 05 |
| F | Give basic applications and types of refrigeration cycles in pharmacy. | 05 |